Computers & Graphics 132 (2025) 104318

Contents lists available at ScienceDirect

Computers & Graphics

journal homepage: www.elsevier.com/locate/cag

Technical Section

Beyond buttons: A user-centric approach to hands-free locomotion in Virtual Reality via voice commands☆

Jan Hombeck *, Henrik Voigt, Kai Lawonn

Friedrich-Schiller-University Jena, Germany

ARTICLE INFO

Keywords:
Virtual Reality
Hands-free navigation
Natural language processing

ABSTRACT

Using speech for hands-free interaction in Virtual Reality (VR) is gaining popularity, supported by advances in natural language processing that enable accurate speech-to-text transcription and intent inference. Prior work has shown that voice-based navigation is effective when hand use is restricted or occupied, particularly when visually salient points in the scene, known as landmarks, can be identified and articulated. However, in many virtual environments, such landmarks may be absent or not easily recognizable, limiting the applicability of voice commands. To address this, we propose three landmark-free, user-centered coordinate systems to support speech-based locomotion in VR: Cartesian, cylindrical, and spherical. Each system uses a distinct encoding and interaction style. The Cartesian system employs a simple three-digit code and requires low cognitive effort. The spherical system prioritizes precision but demands higher mental effort. The cylindrical system combines elements of both, offering a balance between usability and accuracy. We evaluated these systems in a quantitative user study with 24 participants, all with backgrounds in information technology. The study assessed the systems' effectiveness for object positioning and navigation, comparing them to teleportation, the standard non-voice locomotion method in VR. Results show that the Cartesian system enables faster and more intuitive navigation compared to the spherical system. Distinct usage patterns were observed, with users predominantly focusing on the central field of view during navigation. Our findings indicate that user-centered coordinate systems are practical to implement and present a viable alternative for speech-driven, hands-free navigation in VR.

1. Introduction

The use of speech as an interaction technique in virtual reality (VR) applications is becoming increasingly popular [1–3]. One obvious reason is the vast improvement in the quality of processing of spoken and written natural language [4,5]. This is particularly evident in the significant public attention drawn by chatbots like ChatGPT. Advances in natural language processing (NLP) have also made speech technologies interesting as a means of locomotion for VR applications, especially in scenarios where the hands are unavailable because they are used for secondary tasks [2,6,7]. These scenarios, also described as *hands-free*, occur for example, in the medical field in surgical procedures under sterile conditions [8,9]. Other fields, such as architecture and art, utilize voice commands to quickly navigate and manipulate scenes [10], thereby increasing the accessibility of applications and enabling users to participate in virtual exhibitions and museums that they could not physically visit. Voice navigation also provides an alternative means of interaction for individuals for whom the use of physical controls is impractical or infeasible [11,12].

The effectiveness of using voice control for hands-free navigation in environments with distinct visual objects has been well-documented in several studies [2,3,6]. However, many VR applications face challenges when there are no prominent visual objects, commonly referred to as *landmarks*, or when describing these landmarks in natural language is not feasible. The goal of our work is to address this gap by providing a voice-based navigation solution for environments that lack sufficient landmarks. This issue arises in both fundamental environments, such as empty houses or rooms, as well as in specialized environments like those used for visualizing biomedical models [13,14]. In these contexts, the absence of visual landmarks can make it difficult to articulate precise positions, both for navigating oneself and for placing virtual objects or markers in specific locations. A similar challenge arises when an object cannot be clearly articulated, such as in a forest, where identifying the exact tree the user intends to navigate to becomes a

☆ This article was recommended for publication by Prof. N. Pelechano.

* Corresponding author.
E-mail addresses: jan.hombeck@uni-jena.de (J. Hombeck), henrik.voigt@uni-jena.de (H. Voigt), kai.lawonn@uni-jena.de (K. Lawonn).

https://doi.org/10.1016/j.cag.2025.104318
Received 20 December 2024; Received in revised form 15 July 2025; Accepted 18 July 2025
Available online 28 August 2025
0097-8493/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

cumbersome task. In the absence of a robust speech interface and hands-free locomotion technique, this process can become an error-prone and frustrating dialog between the user and the system, leading to a less satisfactory experience. We directly address the challenges of landmark-based voice navigation as a critical step towards enabling truly immersive and versatile VR experiences.

To close this gap in voice-based locomotion, we explore the concept of **user-centered coordinate systems** as a potential solution. Aeronautics [15] serves as a reference field due to its well-established and effective navigation systems. In aeronautics, sophisticated coordinate systems, such as **Cartesian** and **spherical** systems, have been employed to tackle the challenges of navigating complex and dynamic environments from a user-centered perspective. By utilizing user-centered coordinate systems to pinpoint locations in both 2D and 3D, we enable a form of interaction with digital environments that was previously not possible in voice-based locomotion. Hence, we undertake a comparison of user-centered coordinate systems through four distinct tasks: (i) *2D navigation in environments with constant height*, (ii) *3D navigation in spatial environments with different heights*, (iii) *2D object manipulation in environments with constant height*, and (iv) *3D object manipulation in spatial environments with different heights*. Our goal is to explore the practicality of these tasks and identify usage patterns that could further improve hands-free navigation. Our contributions can be summarized as follows:

- Development of a real-time voice control system for **speech-based locomotion and object placement** in environments lacking visual landmarks.
- Comparison of user-centered **Cartesian** and **spherical** coordinate systems for hands-free locomotion in VR .
- Identifying and analyzing **usage patterns** for speech-based locomotion and object placement tasks.
- Offering a **open-source access** to encourage further research and development in the field of voice-based navigation.[1]

2. Related work

In this section, we place our approach in the context of previous work. In Section 2.1, we first refer to influential work that focuses on locomotion in VR, particularly hands-free locomotion techniques. In Section 2.2, we refer to related work in NLP that focuses on speech recognition and language modeling, as these technologies are used in our voice control system.

2.1. Locomotion techniques

2.1.1. VR locomotion techniques

In their survey, Martinez et al. [16] classified VR locomotion techniques into five general categories: walking-based, steering-based, manipulation-based, automated, and selection-based. *Walking-based* locomotion methods use the physical motion of the feet to change position in the VR scene [17]. *Steering-based* locomotion methods impose continuous motion on the user. Controllers are often used to determine the direction of motion [18,19]. Other approaches use hand gestures [20], eye-tracking [19], or leaning [18] to specify the movement direction. *Manipulation-based* locomotion methods use the idea of directly manipulating camera or world coordinates [21]. This can be achieved by using virtual hands [22]. *Automated* locomotion techniques are based on the idea of following a predefined point trajectory in a virtual environment [23]. The user has almost no influence on the execution of this trajectory, but can influence the speed of the tour [24]. *Selection-based* locomotion techniques are fundamentally inspired by the concept of teleportation [25]. In this context, users choose a target location within the virtual environment by either pointing at it [26] or gazing at it [27]. VR can also enhance distance perception and visualization [28,29] compared to desktop applications [30,31]. Among various VR movement methods, teleportation remains the most prevalent.

2.1.2. Hands-free locomotion techniques

Hands-free locomotion techniques, on the other hand, are most often found in the steering or selection-based locomotion categories. Prithul et al. [32] compare eye winks, mouth gestures, and dwell as a location selection technique. Calandra et al. [6] examine speech and gaze in hands-free scenarios in a point-to-point navigation task. The leaning approach, which involves a relative change in position when a user leans in a particular direction, can significantly enhance the perceived sense of self-motion. Research has shown that this technique provides a highly immersive self-motion experience [18,33]. A comparison between leaning and voice-controlled locomotion in sequential navigation [3] has highlighted the convenience and faster time-to-target measures of voice-controlled techniques, making them a more efficient option in certain contexts. Steering-based approaches, on the other hand, utilize methods such as in-place walking [34] and pedal-based steering [35], which translate local foot movements into position changes, thereby enhancing immersive navigation experiences. Tregillus et al. [36] used head tilt approaches for direction determination and walking in place for speed adaptation. Zielasko et al. [37] evaluated the performance of five different hands-free systems and found that the leaning and accelerometer pedal methods performed best. Building on this work, Flemming et al. extended the evaluation by investigating various status control methods for seated leaning approaches [38].

2.1.3. Testbeds for evaluating VR locomotion

When evaluating various locomotion techniques, the predominant focus revolves around key performance metrics [39]. These metrics encompass accuracy [40,41], operational speed [40,42], and error proneness [42,43]. Some evaluations extend their purview to encompass physical aspects [44,45] and mental exertion [44,45]. Additionally, ease of use [43,46] is also considered a significant criterion for assessing the utility of different locomotion techniques. Cannavò et al. [39] presents a locomotion testbed that encompasses a diverse range of locomotion tasks, including multiple movement approaches. Another prominent testbed, known as the Locomotion Usability Test Environment (LUTE) [47], shifts the focus towards the environmental aspects of locomotion. This approach provides a nuanced perspective on locomotion challenges across varying terrains and distances.

2.1.4. Coordinate systems for navigation

Coordinate systems have been extensively studied for their applications in aeronautics [15] and satellite navigation [48], as well as in computer interaction, particularly where the use of a mouse and keyboard is unfeasible. Among these applications is the area of scanning systems utilized by nonspeaking individuals with severe motor disabilities. Biswas et al. [49] introduced eight-directional scanning, cluster scanning, and block scanning systems to assist people with motor disabilities in interacting with computers. Similarly, Blackstien-Adler [50] explored approaches using Cartesian and polar coordinate systems. While the goal of improving accessibility and reducing the mental and physical effort for selection-based tasks is shared between our work and traditional scanning systems [51], there are significant distinctions between the two. Traditional scanning systems rely on visual selection with minimal physical input and rely heavily on eye-gaze [52] or single-switch [53] activation. Our approach utilizes voice commands as a means of human–computer interaction. Additionally, scanning systems are typically employed for fundamental 2D computer interactions, such as browsing the web [54], whereas our system is

[1] https://github.com/jhombeck/Beyond-Buttons—Hands-Free-Locomotion-in-VR-

specifically designed for 3D navigation tasks in VR environments without landmarks. Although the overarching objectives are similar, the methods of interaction, the technology used, and the dimensionality of the environments significantly differentiate our approach from conventional scanning systems. Wang et al. [55] investigated the use of coordinate systems from both allocentric and egocentric perspectives to evaluate spatial learning in mixed reality environments. Their approach aimed to enhance users' spatial awareness, and results indicated that participants exhibited improved recall of landmark positions. While their findings demonstrate that coordinate systems can support better recall of spatial relationships, our work is clearly distinguished by its focus on utilizing coordinate systems for voice-based navigation. Nonetheless, Wang et al. effectively highlighted the potential benefits that coordinate systems can offer in enhancing spatial understanding.

2.2. Language technologies and their VR applications

2.2.1. Speech recognition

Reliable speech-to-text transcription is essential for successful speech-based interaction in VR. Recent developments in NLP have greatly increased the accuracy of these systems and also improved their real-time applicability [56]. The *wav2vec* approach, and its successor *wav2vec 2.0* [57], has been a important development for enabling speech-to-text transcription. There are several ways to optimize the performance of this approach by using knowledge distillation [58] or weight quantization [59]. *Whisper's* state-of-the-art transformer-based architecture uses even more extensive pre-training than *wav2vec* to deliver state-of-the-art accurate speech-to-text transcriptions for multiple languages [5].

2.2.2. Language modeling

The advent of large language models has dramatically improved the ability of machines to understand and infer communication intent from human speech. Open-source models such as FLAN-T5 [60], Cerebras-GPT [61], or Llama [62] enable their application even on commodity hardware. The paradigm of fine-tuning large pre-trained language models to diverse instructions and/or dialog data is shown to be an effective means of generalizing to unseen tasks and inferring their solution from context [63]. To make these models run on computationally weak hardware, knowledge distillation [64], quantization [65], and fast inference frameworks such as *ONNX* [66] are primarily used. The state-of-the-art depends on specific requirements, including whether the system prioritizes high accuracy or speed, and whether it is designed to be open-source and free or proprietary and paid.

2.2.3. VR voice applications

A large number of voice control systems implement *command interpreters*. In this approach, the user's speech input is transcribed and compared to a predefined set of commands, triggering the corresponding action. This method is widely used in various applications, such as voice-guided implant positioning for surgical planning [67]. In VR environments, this approach is commonly found in works where inputs like gestures and speech are combined to manipulate objects [68] or to compose individual scenes [69]. More advanced systems rely on *dialog interfaces*, where users interact in conversational exchanges. For instance, Chilufya et al. [70] utilize a virtual receptionist to guide students around campus. Medical training applications allow physicians to rehearse therapeutic conversations in VR [71]. Ferracani et al. [11] have demonstrated voice-controlled locomotion for people with motor disabilities, enabling them to take virtual museum tours. Furthermore, Hombeck et al. [3,72] used LLM approaches to investigate the performance of voice-controlled locomotion across various tasks. Our work builds upon this by extending voice control to environments devoid of visual landmarks, further expanding its applicability.

3. Methods

This section offers an overview of the various methods compared in this study. In Section 3.1, we investigate the distinct properties of *Cartesian*, *cylindrical*, and *spherical* coordinate systems. Section 3.2 describes the NLP architecture employed for recognizing speech commands.

3.1. User-centered coordinate systems

We are comparing three coordinate systems for voice-controlled locomotion: **Cartesian**, **cylindrical**, and **spherical**. All systems are inspired by concepts from the field of aeronautics [15], but their design, interaction methods, and use cases are tailored to our specific approach. In our study, these systems are applied to movements in flat environments, changes in spatial positions, and the placement of objects. The coordinate system was always aligned with the user's view direction, meaning that if the user turned to the left, the system would immediately follow and remain centered within the user's field of view. Several core principles were considered in the design of these locomotion systems to ensure usability and precision. Each system was developed based on distinct guiding principles. The *cylindrical* system, however, represents a hybrid approach that combines elements from both the *Cartesian* and *spherical* navigation paradigms.

Usability and simplicity. The *cartesian* coordinate system aims to provide a user-friendly interaction with minimal cognitive load. To achieve this, we implemented a single-number encoding, enabling users to specify a position using a single number. This simplicity is crucial for minimizing user errors and improving the overall experience. This approach is supported by Sweller's cognitive load theory [73] and Mayer's principles for reducing cognitive load [74].

Precision and flexibility. The *spherical* coordinate system employs an encoding method specifically designed to enhance precision. This system utilizes longitude, latitude, and distance values to achieve floating-point accuracy. It was specifically developed for scenarios that require more accurate navigation, striking a balance between usability and functionality. Research by Mine et al. [75] and LaViola et al. [76] highlights that precise selection techniques enhance user interaction by minimizing errors and speeding up task completion.

User interface. As voice input is the primary mode of interaction with our system, the user interface is intentionally minimalist. When the user presses the trigger on the left-hand controller, the microphone is activated and begins recording. Upon release, the captured voice input is processed. To provide clear feedback, a small sphere is attached to the digital representation of the user's left hand. This sphere appears green while the microphone is active and red when it is inactive, ensuring that the user is aware of the microphone's status and increasing confidence in the system's responsiveness. Additionally, the recognized voice is displayed as text on the digital left hand. This feature helps to reduce user frustration in cases where speech input may be misunderstood by making the system's transcription of the command visible. The annotations are always attached to the digital left hand and move in real time with the user's hand, ensuring that feedback remains visible and contextually relevant during interaction.

3.1.1. Cartesian coordinate systems

In Fig. 1, we present an example of our *Cartesian* coordinate system design. This design features rectangular grids that are generated in front of the user's position, synchronously updating as they move. Grid elements can be labeled with: (a) *numbers*, (b) *characters*, or (c) *hybrid* combinations of both, akin to the labeling of chessboards. In our case, we choose to use only numerical values. The grid coordinates range from −999 to 999, with the z–x plane spanning from 0 to 99. The y-axis represents height, with values in the hundreds of digits. This setup enables users to intuitively encode changes in their position, associating

two-digit numbers, such as 12, with shifts in the z–x plane. Using the hundreds digit, like 123, elevates the user by one unit in the y-direction and positions them in grid 23 of the z–x plane. The distinction between two-digit and three-digit numbers was intentional, as locomotion is often only two-dimensional. In such cases, specifying three digits, where the first digit would always be zero, is unnecessary. To reduce effort, users can simply articulate "23" instead of "023" when only 2D locomotion is required. Negative numbers, e.g., −156, move the user down one unit in the y-direction and to the 2D horizontal grid 56. Due to the decision to use negative numbers to represent height, we are unable to assign negative values for left and right movement, as this approach is designed to encode locomotion using a single numerical value. Consequently, the user's central reference positions are defined as 004 and 005. The grids are sorted in local *yzx* order relative to the user, facilitating quick location changes by memory. For instance, the grids 004 and 005 represent the units directly in front of the user. This logical arrangement ensures that the grids align with the user as the origin of the local coordinate system for navigation. While the step size between grid points can be customized, we have selected a default distance of 1 meter. This setting can be adjusted in real time to create finer or larger steps, thereby modifying the overall dimensions of the grid. However, for the study, we chose to limit the default distance between grid points to 1 meter to align the distances used in the *cylindrical* and *spherical* approaches, ensuring the comparability of results. The color and opacity of the grids are customizable based on the background. As the default, we utilize a light blue color with an opacity of 0.5. This choice ensures that the grids stand out conspicuously without completely obscuring the surrounding environment.

Implementation. Within the *Cartesian* system, each location is represented as a single plane featuring a slightly transparent texture overlaid with a text field. When the system is activated, 100 such tiles are initialized and arranged in a compact grid in front of the user. These tiles are then numbered according to our encoding scheme. Once a voice command is recognized and a target number is identified, the system locates the corresponding tile and selects its center as the user's new destination. The Y-Value Tower is implemented in a similar manner but differs in spatial configuration. Here, the planes are arranged vertically in a straight upward formation, with each plane oriented to face the user.

3.1.2. Spherical coordinate systems

In Fig. 3(b), we present an illustration of our *spherical* coordinate approach. The spheres are centered around the user, and their positions are defined by a *longitude (lon)* angle ranging from −180° to 180° , a *latitude (lat)* angle between −90° and 90° , and a *radius*. This design enables users to pinpoint a specific point in 3D space by expressing the *spherical* coordinates as 2D lon-lat coordinates, with the distance represented by the radius. If only a single angle is specified without additional markers, it defaults to the lon, facilitating 2D *spherical* navigation. In cases where no radius is specified, it defaults to a radius of one. When used solely with distance and longitude values as a 2D locomotion technique, the spherical system effectively represents the use of a polar coordinate system. The visual representation of the 2D *spherical* system consists of multiple circles based on different radii, with a default distance of 1 m between each circle. Longitude angles are labeled in 10° increments at a 5 m distance. Upon entering 3D spherical navigation, an additional circle appears at the 0° longitude position, oriented along the latitude direction. Here, extra labels are added at 10° intervals. While the labels are placed at 10° intervals, users enjoy the flexibility to freely select any value within the range of the minimum and maximum possible angles as their input. Similar to the grid, sphere coloring adheres to the principles described earlier. Users can adjust the degree intervals in both the lon and lat directions of the sphere according to their preferences.

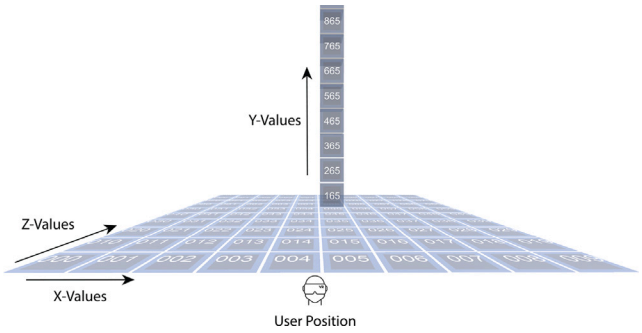
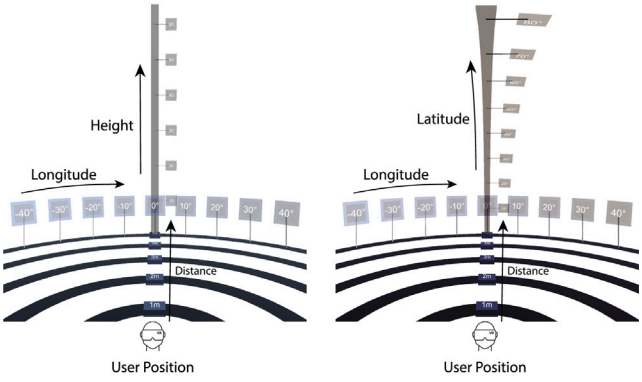
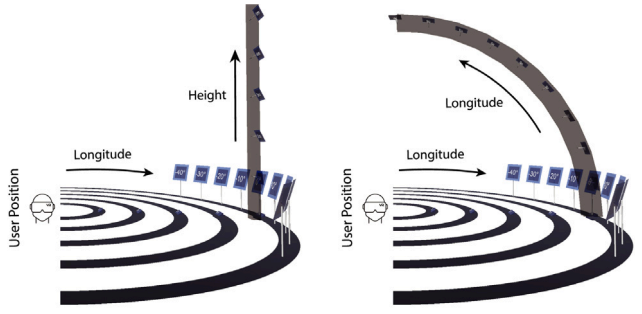


Fig. 1. An overview of the Cartesian coordinate system in a zoomed-out format is provided to illustrate its three-dimensional structure. The user is positioned between the coordinates four and five.



(a) Cylindrical coordinate system. (b) Spherical coordinate system.

Fig. 2. An overview of the *(a)* cylindrical and *(b)* spherical navigation systems is presented from the user's perspective. The primary distinction between the two systems lies in their encoding of height. In the cylindrical system, height is represented as a linear vertical displacement. In contrast, the spherical system encodes height using a latitude angle, which specifies the vertical position on the surface of a sphere.



(a) Cylindrical coordinate system. (b) Spherical coordinate system.

Fig. 3. An overview of the *(a)* cylindrical and *(b)* spherical navigation systems is presented as view from the side.

Implementation. The foundation of the *spherical* system is based on black isolines, which are rendered within the fragment shader and mapped onto an additional plane positioned above the actual ground surface. Any region outside the isoline boundaries is rendered transparent. Supportive elements, such as longitude and latitude labels, are incorporated into a single prefab and positioned at the appropriate angle along the 5-meter isoline. As the user moves and rotates their head, the prefab, including the isoline plane and all associated objects required for the technique, moves and rotates in accordance with the user's head orientation. The rotation of the prefab is constrained to the

Y-axis, ensuring that it remains horizontally aligned and consistently flat on the ground.

3.1.3. Cylindrical coordinate systems

In Fig. 2(a), we present an illustration of our *cylindrical* coordinate approach. This system represents a hybrid between *Cartesian* and *spherical* coordinate systems. This fusion allows for intuitive navigation and positioning in a cylindrical reference frame. The system is centered around the user and defines positions using a *longitude (lon)* angle ranging from −180° to 180° , a forward *distance* extending radially from the user, and a *height* component that extends vertically, either upward or downward along the y-axis. In this representation, spatial locations in 3D are encoded using a combination of angular direction and a compact numerical encoding of distance and height. For instance, a label such as "35" denotes a point located 3 meters upward and 5 meters forward in the direction specified by the longitude angle. This notation enables an intuitive and compact way to specify positions within the cylindrical frame, while still providing accurate positioning options. The visual layout of the 2D cylindrical system consists of concentric radial lines centering around the user, with angular intervals labeled every 10° and distance markings placed in one meter increments. In the 3D cylindrical mode, a tower indicating the height at one meter intervals.

Implementation. The *cylindrical* system is implemented in a manner similar to the *spherical* system but differs in its representation of height information. Instead of rendering a spherical slice, it employs a linear vertical height encoding. Once a voice command is processed, the system first identifies the corresponding position on the ground plane and then elevates the user vertically based on the tens digit of the input number.

3.1.4. Teleportation

To establish a baseline for comparison with the voice-based approaches, we included on of the most commonly used navigation method in VR, teleportation, into our evaluation [18]. It is important to note that, unlike the voice-based systems, teleportation still relies on physical controllers and cannot be considered a hands-free approach. It is primarily used as a comparison point in this work. In the teleportation system, users initiate the teleportation by pressing the touchpad on the controller. While the button is held down, a parabolic line is projected from the controller, showing a preview of the teleport destination. This line intersects with the environment, turning green when aimed at a valid location and red when directed at an invalid area. Upon releasing the button, teleportation is initiated and the screen fades to black. The user is then teleported to the point where the line intersects with the environment, after which the screen fades back in. The same locomotion method is also employed for all voice-based approaches. Once the desired destination is determined, the screen fades to black, teleportation is performed, and the screen fades back in. This approach is intended to prevent sudden movements that may induce motion sickness. The implementation is based on the default teleportation functionality provided by SteamVR.

3.2. Voice control system

This section examines the voice control system, detailing its architecture and the commands it supports. Unlike traditional systems that rely on specific keywords, our system uses semantic mapping to interpret user intentions, allowing various phrasings.

3.2.1. System architecture

Our voice control system comprises two main components: a speech-to-text module and a language model module. Both are implemented as services within a FastAPI server, which runs in the background inside a Docker container. The Unity front-end application [77] communicates with these services via web requests. To capture user speech, we continuously record audio using the Valve Index microphone at a sampling rate of 16 kHz. The raw audio stream is directed to the speech-to-text service on the server. To delineate the start and end of a voice command, users could toggle the microphone using a button to mute or unmute it. The recorded WAV audio file is then processed by a *Whisper* [5] speech-to-text transcription model. We utilize the *small* version of Whisper with 8-bit quantization to ensure real-time transcription performance. Once transcription is complete, the resulting text string is forwarded to the language model service. Within this service, a conversation manager processes the dialog. It constructs a prompt that incorporates a set of callable functions, either predefined or dynamically added at runtime. Each callable function must include a docstring describing its signature, parameters, and purpose. The transcribed user query is embedded into this prompt and sent to a *Gemini 2.0 Flash* language model instance via Google Cloud Vertex AI for text completion. Based on the input query, the language model determines the appropriate function to invoke and generates a response in the form of a function call with the correct arguments, enabling seamless execution of voice-driven commands. This string is then sent back to the Unity front-end, where the function is finally executed. Our code and a detailed architecture description are available as an open-source repository at <*anonymized link*>.

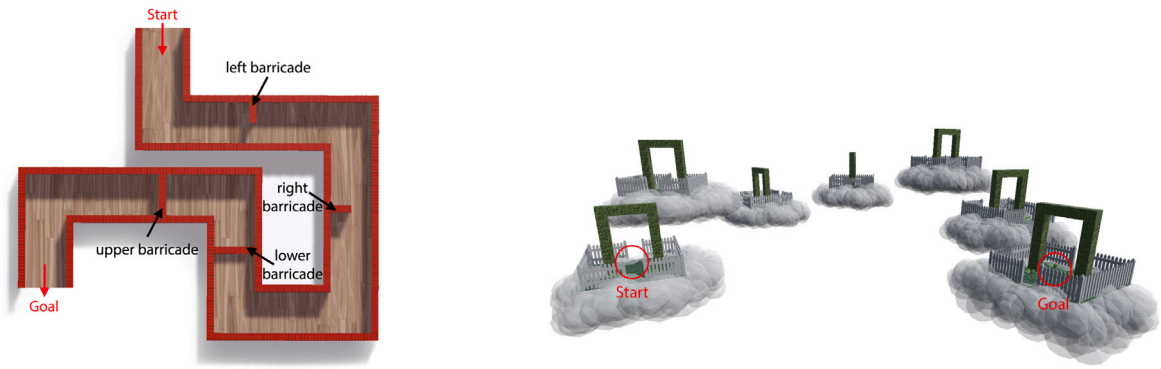
3.2.2. Voice command types

Our voice control system implements three different types of locomotion commands: (i) absolute positioning commands, (ii) relative positioning commands, and (iii) specific commands.

Absolute positioning commands. Absolute positioning commands allow you to specify a desired position in the *Cartesian*, *cylindrical*, and *spherical* coordinate system. To change the position on the *Cartesian* grid, the user simply specifies the desired position by the grid number, such as '*Go to grid* 544'. After the speech-to-text transcription, the language model identifies the correct function and its parameter and initiates the execution of the function call `move_player_grid(544)` in the Unity engine. After execution, the user's position will be adjusted to the `544th` grid. Similarly with the *cylindrical* and *spherical* coordinate system. In the *spherical* approach the user specifies a location change in longitude, latitude, and radius as follows '*Move to 30 degrees longitude, 15 degrees latitude, and 5 meters*'. The transcribed audio is interpreted by the language model and the correct function is called with the parameter list `30, 15, 5` ending in the function call `move_player_sphere(30,15,5)`.

Relative positioning commands. Relative positioning commands offer a flexible way to specify changes in position or orientation based on the user's current context and are applicable across all navigation techniques, regardless of the underlying coordinate system. These commands can be used for self-locomotion tasks, modifying the user's own position or orientation, as well as for object placement tasks, where an object's position or rotation is adjusted relative to its current state. For example, a command such as "*Move one step to the right*" can be interpreted as `move_player_relative('Right', 1)` and moves the player relatively to its own position. Similarly, orientation adjustments, such as "*Turn the sofa 45 degrees*", are interpreted as `rotate_object('Sofa', 45)`, with rotation applied around the Y-axis by default. This approach enables users to adjust both their own position and the position of virtual objects in the environment more effectively and intuitively.

Special commands. Special commands are mainly used to trigger additional functions, such as building visual aids like grid towers ('*Show me a tower at 85*' calls `build_grid_tower(85)`) for better estimation of height differences in *Cartesian* coordinate systems, and activating latitude slices in spheres for better estimation of elevation angles ('*Show me sphere slice at 20 degrees*' calls `show_sphere_slice(20)`). A visual representation of this is shown in Figs. 1, 2 and in the lower row of Fig. 5.



(a) Top-down view of the 2D maze

(b) Side-view of the 3D maze

Fig. 4. Participants face challenges involving two distinct maze tasks. The first task (a) centers on 2D navigation, restricting movement to the XY plane featuring a maze with nine turns. In contrast, the second task (b) introduces a 3D navigation element, comprising seven platforms that incorporate varying heights into the navigation process.

4. User study

Our study design draws inspiration from established locomotion testbeds [39,47]. They emphasize the significance of both 2D and 3D movements for creating a user-friendly locomotion technique. Expanding on this, our study further incorporates speech-based object manipulation [78,79]. In line with these influences, we opted to adopt a recommended maze setup [3,80], which includes both medium and short distances for two-dimensional traversal (see Fig. 5 top row). Additionally, we introduced a three-dimensional maze (see Fig. 5 bottom row) incorporating height as an additional challenge.

4.1. Research questions

Based on the open challenges in the area of landmark-free locomotion, we formulated the following research questions for investigation:

- **[Rq1]** **Locomotion Tasks**: How do interaction and user feedback differ between self locomotion and object manipulation?
- **[Rq2]** **Coordinate System**: How do Cartesian, cylindrical and spherical approaches differ in terms of usability and performance?
- **[Rq3]** **Interaction Modality**: How does the performance and usability of voice-based interaction compare to non-voice-based interaction methods?
- **[Rq4]** **Interaction Patterns**: What interaction patterns can be observed in landmark free speech-based locomotion systems?
- **[Rq5]** **Voice Locomotion**: How do voice commands support locomotion in environments without landmarks?

4.2. Study setup

To ensure the comparability of study results, we conducted the experiment in a controlled, in-person, in-lab environment. Before starting the study, we conducted a trial to ensure that all participants understood the tasks and commands. This training phase lasted a maximum of five minutes, during which we addressed any questions. During the study, only the participant and the research staff were present to minimize external influences. To further mitigate unintended voice commands, participants could mute and unmute the microphone by pressing a button on the left-hand controller. Upon completing the study, each participant was asked to complete a questionnaire and participate in a brief debriefing session. On average, the entire process took approximately 60 min per participant, depending on individual speed and engagement. The study was developed using the Unity framework [77]. We employed a system consisting of an Intel(R) Core i9-12900K processor, an Nvidia RTX 3090 graphics card, 32 GB of RAM, and Windows 11. This configuration allowed us to run our VR setup and voice control system without encountering any performance issues. For the HMD, we opted for the Valve Index.

4.3. Procedure

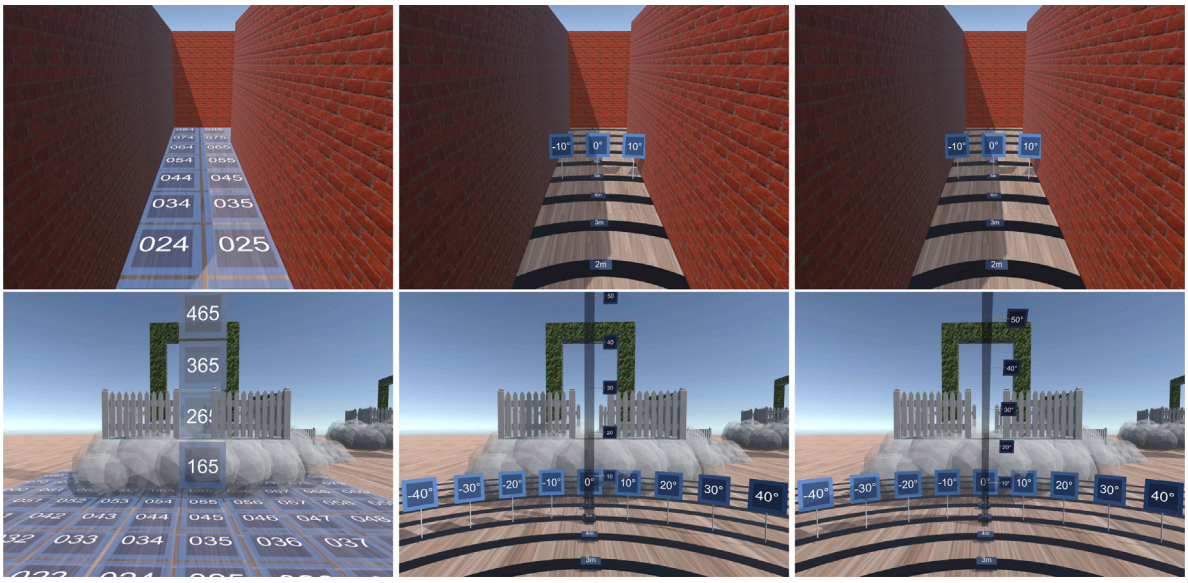
The study consisted of four main tasks: (1) 2D navigation, (2) 3D navigation, (3) 2D object manipulation, and (4) 3D object manipulation. Each task was performed a total of four times, once for each of the locomotion techniques: the Cartesian, cylindrical, and spherical voice-based approaches, as well as teleportation, which served as the non-voice-based baseline technique.

2D navigation. We utilized a simplified maze to evaluate the navigation capabilities of voice-controlled locomotion. Participants were tasked with navigating a maze (see Fig. 4(a)) consisting of a predefined path that involved left and right turns. There was no wrong turn within the maze. The maze included four barriers that partially blocked the user's view to the next turn, adding visual clutter to better mimic real-world navigation challenges. The barriers were placed on the left, right, top, and bottom sides of the maze. The maze was designed so that each turn could still be reached with a single voice command, regardless of the navigation system used. Thus, to complete the maze optimally, at least seven voice commands were required: six for the turns and one to reach the goal.

3D navigation. One limitation of previous voice locomotion techniques is the inability to navigate across different height levels. Therefore, we constructed a three-dimensional environment with multiple platforms suspended in the clouds (see Fig. 4(b)). Each of the seven platforms was uniquely positioned at different locations with different heights. As a result, participants had to navigate upwards and downwards three times each throughout the experiment. The distance between each platform was designed to be reachable with a single voice command, regardless of the navigation system employed. Completing this task optimally required at least six voice commands (seven platforms, excluding the starting platform).

Object manipulation. For the task of object manipulation, each participant was presented with a miniature model of a room containing six distinct furniture items, along with a full-sized, empty room (see Fig. 6). The goal was to accurately and efficiently recreate the modeled environment. Certain objects, such as the `bed`, required only correct placement within the scene. Others, including the `sofa` and `desk`, also needed to be rotated using voice commands. Additionally, items like the `lamp`, `vase`, and `mirror` featured distinct height levels, introducing further complexity to their placement. To complete the task effectively, participants required a minimum of eight voice commands: six for placing each object and two for the required rotations.

Study order and counterbalancing. To ensure a progressively increasing level of difficulty, the study tasks were carefully ordered, beginning with the simplest and advancing to more complex ones. The sequence



(a) Cartesian (b) Cylindrical (c) Spherical

Fig. 5. Examples of study tasks as perceived by participants: *(a)* Cartesian, *(b)* Cylindrical, *(c)* Spherical. The top row shows the 2D Navigation task : The objective is to traverse a maze. the bottom row shows the 3D Navigation: The objective is to successfully navigate a 3D environment consisting of various platforms.

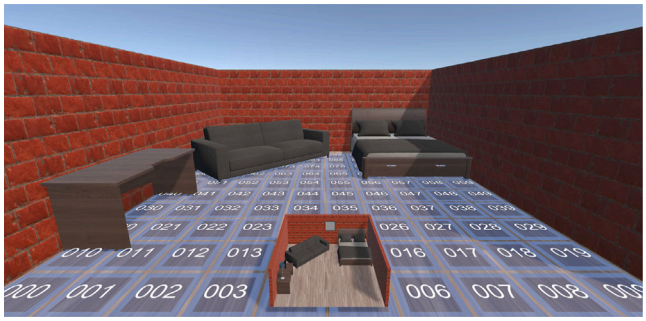


Fig. 6. 2D/3D Object Manipulation task. The objective is to accurately replicate a provided miniature room layout. The example shown uses the *Cartesian* coordinate system. The 2D objects are already positioned, and the 3D objects are about to be placed.

started with 2D Navigation, followed by 3D Navigation, then 2D Object Manipulation, and concluded with 3D Object Manipulation. Each task was performed using four locomotion techniques: three voice-based methods: Cartesian, Cylindrical, and Spherical, and the commonly used teleportation technique, which served as a baseline. The study was conducted sequentially for each technique, with participants completing all tasks for one technique before proceeding to the next. To mitigate order effects, the study employed a counterbalanced design in which the presentation order of the four locomotion techniques varied across participants. Given four techniques, there are 24 possible permutations of presentation order. We utilized all permutations to determine the task order, ensuring that each configuration was evenly distributed among participants and thereby minimizing potential order effects.

4.4. Participants

In our study, we included a group of 24 participants, comprising 19 males and 5 females, with ages ranging from 21 to 32, with an average age of 25. Participants were recruited from within our faculty through voluntary participation. Recruitment was conducted via internal announcements and direct invitations. Participation was entirely optional, and no monetary compensation or other incentives were offered. All participants were informed about the purpose of the study and provided informed consent prior to taking part. Educational backgrounds among participants varied, with 8 having completed a master's program, 14 holding a bachelor's degree, and 2 possessing a Ph.D. Regarding VR usage, 8 participants had never used VR before, 6 had used it once or twice, 9 reported using it a few times per year, and 1 participant used VR more than a few times per month. All participants had a background in information technology, either as software engineers or currently engaged in research or as students within a related field.

4.5. Measures

The following section outlines the types of measures employed in our study, which are divided into objective and subjective categories. Subjective measures were collected through a post-study questionnaire administered via a Google Forms document. These measures capture participants' self-reported experiences, preferences, and perceptions related to the interaction techniques. Objective measures were recorded in real time within the Unity environment during task execution. Each objective metric was computed or automatically captured at runtime, enabling precise and consistent data collection without user intervention.

In our study, the independent variables include the interaction modality (*voice-based* vs. *teleportation*), the coordinate system used for locomotion (*Cartesian*, *cylindrical*, and *spherical*), and the task type (*2D navigation*, *3D navigation*, *2D object manipulation*, and *3D object manipulation*). The dependent variables comprise both objective and subjective measures. Objective metrics include *task completion time*, *number of voice commands* issued, and spatial *accuracy*. Subjective metrics were obtained through standardized questionnaires and assess *sickness*, *comfort*, *presence*, and *usability*.

4.5.1. Objective measurements

In our evaluation, we have incorporated three objective measurements. The first commonly used metric in evaluating locomotion systems is the *time to target* [3,16,18], which measures the time participants require to complete a given task. Additionally, we are interested in the *number of voice commands* needed to accomplish the task. To gain a better understanding of the participants' *interaction pattern*, we have recorded the corresponding target locations including grid numbers

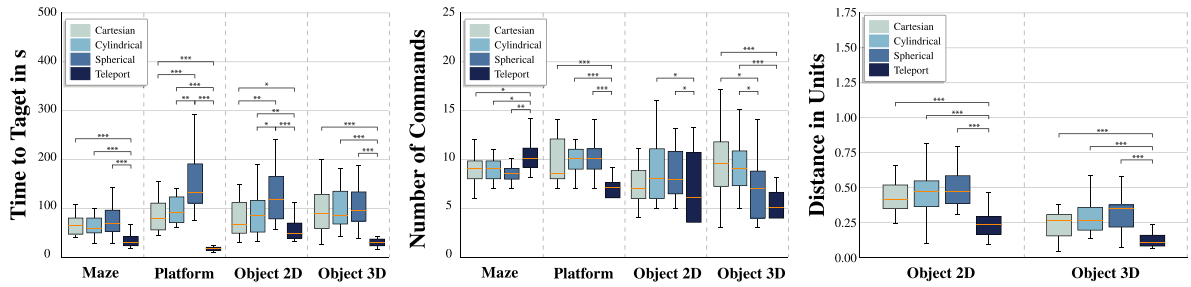


Fig. 7. Comparison of user-centered Cartesian and spherical coordinate systems in terms of *left:* time to target in seconds, *middle:* number of voice commands required to complete the task, *right:* distances offset of objects to be placed.

or longitude/latitude values. For tasks involving object manipulation, *accuracy* is an important factor [40,41,81]. Therefore, we recorded the Euclidean distance error in units of measurement. A small distance from the gold standard results in a high *accuracy*.

4.5.2. Subjective measurements

The questionnaires chosen as subjective measures are generally validated and widely used. However, in some cases, they may not fully align with our focus on speech-based locomotion. We adapted the questionnaires to better suit our research needs. For example, the "Device Assessment Questionnaire (DAQ)" includes questions on physical fatigue related to finger, wrist, arm, and shoulder movements, which are irrelevant in our hands-free navigation context and could lead to misleading results. This streamlined approach ensures that we focus on questions most relevant to assessing locomotion tasks, in line with previous studies [3,18] and recommendations [82]. We have documented the selection process for each questionnaire to provide clarity and context for our findings, striking a balance between our study's unique requirements and the need to align with existing literature. All questions are answered using either a 5-point or 7-point Likert scale, depending on the recommended format specified by each questionnaire.

Sickness. One of the most crucial criteria for user-centered applications is to ensure that users are not experiencing any sickness within the environment. Regardless of how accurate and fast an application may be, if it causes discomfort, it is unlikely to find practical application. Therefore, we have incorporated questions 1, 3, 4, and 8 from the Simulator Sickness Questionnaire (SSQ) [83]. These questions specifically address aspects such as *general discomfort*, *headache*, *eye strain*, and *nausea*.

Comfort. While a high sickness score may serve as an indicator against adopting the system, ensuring user comfort is essential for evaluating the convenience of a system. A higher comfort score indicates a greater likelihood that users will embrace the system. We have adapted the Device Assessment Questionnaire (DAQ) [84], incorporating questions 2, 3, 4, 5, and 6. These questions encompass various aspects such as *overall comfort*, *ease of use*, *accuracy*, and *physical* and *mental exertion*.

Presence. In the context of virtual environments, achieving a sense of presence or immersion is essential for fully engaging users in the digital world. To evaluate the level of immersion across different locomotion systems, we use the Igroup Presence Questionnaire (IPQ) [85]. We selected all questions from the IPQ, which cover various aspects of presence that can be categorized into *general presence*, *spatial presence*, *involvement*, and *experienced realism*. The work of Melo et al. [86] investigated the thresholds at which presence scores can be considered acceptable. We utilize their grading framework to contextualize our results in relation to a broad range of applications that employ the IPQ for evaluating presence. Melo et al. categorized presence scores into letter grades ranging from A to F, A being the highest.

Usability. The usability of the system is important for an intuitive user experience. To assess this, we employed the System Usability Scale (SUS) [87], which consists of ten questions. These questions cover various aspects of usability and have been categorized into the following: *reusability*, *complexity*, *integration*, *consistency*, *learnability*, and *confidence*.

5. Results

The following section presents the results of our study. Objective measurements are illustrated in Figs. 7, 8, 10, 9, and 11. Subjective measurements are depicted in Figs. 12, 14, 13, and 15. To assess differences between techniques within each subcategory, pairwise comparisons were conducted using Welch's t-test, which does not assume equal variances between groups. This test was chosen for its robustness to heteroscedasticity. Statistical significance was determined based on unadjusted p-values, with thresholds of $p < 0.05$, $p < 0.01$, and $p < 0.001$. For variables measured on an ordinal scale, such as Likert-type responses, we employed non-parametric statistical tests that do not assume interval-level measurement or normal distribution. Specifically, the Mann–Whitney U test was used for pairwise comparisons between independent groups. These results are included in our figures using asterisk notation to indicate statistical significance, with * representing $p < 0.05$, ** representing $p < 0.01$, and *** representing $p < 0.001$. In our results, we emphasize the mean values to highlight central tendencies. For a more detailed analysis, including standard errors, we recommend referring to the corresponding figures.

5.1. Objective measures

5.1.1. Time to target

In 2D navigation tasks, the *Cartesian* system demonstrated the best performance with an average task completion time of 67 s, outperforming the *cylindrical* system (70 s) and the *spherical* system (80 s). *Teleportation* resulted in the fastest completion time at 34 s. Similarly, in 3D navigation, the *Cartesian* approach again showed superior efficiency with an average time of 87 s, compared to the *cylindrical* (101 s) and *spherical* (155 s) approaches. *Teleportation* remained the fastest method, with a completion time of 17 s. For 2D object manipulation, the *Cartesian* system (79 s) also outperformed the *cylindrical* (89 s) and *spherical* (124 s) systems. *Teleportation* was the most efficient, completing tasks in just 30 s. In 3D object manipulation tasks, a similar trend was observed. However, the *cylindrical* approach (97 s) outperformed both the *spherical* (101 s) and *Cartesian* (104 s) systems in terms of average completion time. *Teleportation* again proved to be the fastest, with an average time of 30 s.

5.1.2. Voice commands

The number of successful voice commands used to complete tasks, as shown in Fig. 7 (center), indicates only minor differences among the voice-based techniques. For 2D navigation, all voice systems required an average of 9 voice commands, while *teleportation* required

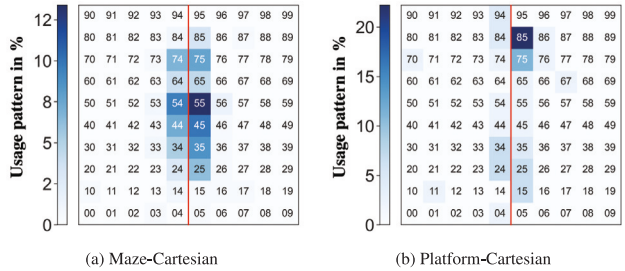


Fig. 8. Distribution of target numbers in the Maze-Cartesian and Platform-Cartesian tasks. Participants were positioned between targets 4 and 5. Color intensity indicates percentage selection per grid cell. The red line denotes the center of the field of view.

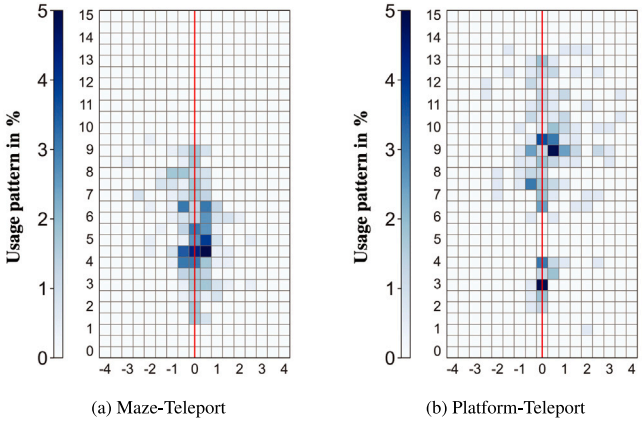


Fig. 9. Distribution of teleportation targets in the Maze-Teleport and Platform-Teleport tasks. Color intensity represents the percentage of teleport selections. The Y-axis shows distance moved; the X-axis reflects lateral movement. The red line marks the field-of-view center.

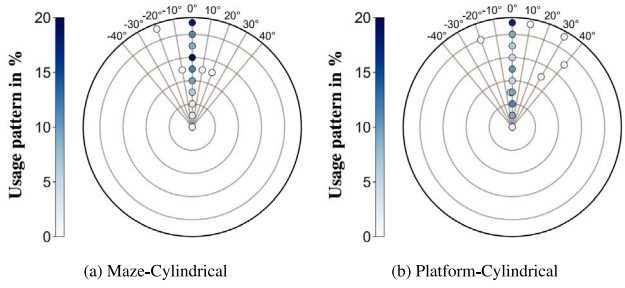


Fig. 10. Distribution of target angles and distances in the Maze-Cylindrical and Platform-Cylindrical tasks, visualized in polar coordinates. Color intensity indicates percentage distribution; each ring spans 2 m. The 0-degree mark aligns with the center of the participant's field of view.

10 inputs on average to complete the task. For 3D navigation, the average number of voice commands increased slightly to 10 for both the *cylindrical* and *spherical* approaches, while remaining similar for the *Cartesian* approach at 9. The *teleportation* technique required fewer inputs, averaging 8. In the context of 2D object manipulation, participants used approximately 8 voice commands on average across all voice-based systems. The *teleportation* method required fewer inputs, with an average of 7. During 3D object placement, the average number of voice commands increased to approximately 9 for both the *Cartesian* and *cylindrical* systems, while remaining at 7 for the *spherical* approach. The *teleportation* technique again required the fewest inputs, averaging around 5 voice commands to complete the task.

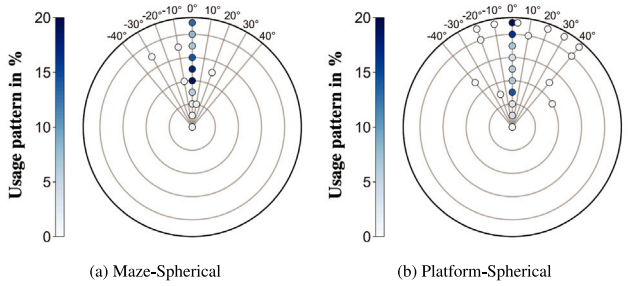


Fig. 11. Distribution of target angles and distances in the Maze-Spherical and Platform-Spherical tasks, visualized in polar coordinates. Color intensity represents percentage distributions; each ring spans 2 m. The 0-degree mark aligns with the participant's central field of view.

5.1.3. Accuracy

For 2D object placement, the average distance offset was 45 cm for the *Cartesian* system, 49 cm for the *cylindrical* system, and 53 cm for the *spherical* system, while the *teleportation* approach yielded a lower offset of 29 cm. In 3D placement tasks, the average distance offset decreased across all systems: 27 cm for the *Cartesian* system, 31 cm for the *cylindrical* system, and 33 cm for the *spherical* system. The *teleportation* approach again achieved the lowest offset, averaging 18 cm.

5.1.4. Interaction patterns

Our results, depicted in Figs. 8, 10, 9, and 11, shed light on the interaction patterns observed during the study. Participants predominantly relied on voice commands to navigate in the direction directly in front of them. For the *Cartesian* system, irrespective of whether it was 2D or 3D navigation, we observed a distinct preference for numbers ending in 4 and 5, which correspond to the values directly to the left and right of the view direction. In the case of the *cylindrical* and *spherical* systems, regardless of 2D or 3D navigation, we found that the most commonly used value was 0 degrees longitude. During the 2D navigation task, participants rarely utilized values outside the range of $-30°$ to $30°$, which align with directions straight ahead or slightly to the right or left of the view direction. In contrast, for 3D navigation, despite the option to specify a longitude angle, almost no participant deviated from the 0-degree longitude. For the *teleportation* approach, a similar behavior was observed as with the voice-based techniques. Almost all teleportation commands were issued within a range of 1 meter to either the left or right of the user, with a slightly higher frequency recorded on the right side compared to the left.

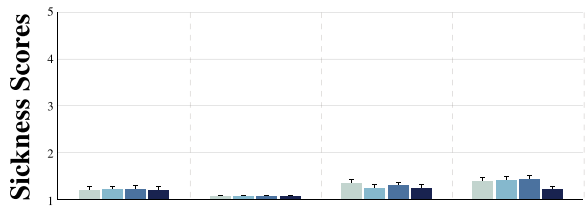
5.1.5. System performance and accuracy

In total, we recorded 2907 voice commands throughout our study. Of these, 28 commands failed, meaning that although a function was successfully identified, it was executed with incorrect parameters. Additionally, our study showed that in 317 out of the 2907 cases, no valid function could be derived from the given voice command. This results in an overall system accuracy of 89%. However, the accuracy varied across the different voice-based approaches. The *Cartesian* system achieved the highest accuracy at 96%, followed by the *cylindrical* approach at 90%, and the *spherical* approach at 80%. The average time required to process and execute a command after it was articulated was 0.61 s. Specifically, the *Cartesian* system required 0.58 s on average, while both the *cylindrical* and *spherical* systems required 0.62 s.

5.2. Subjective measurements

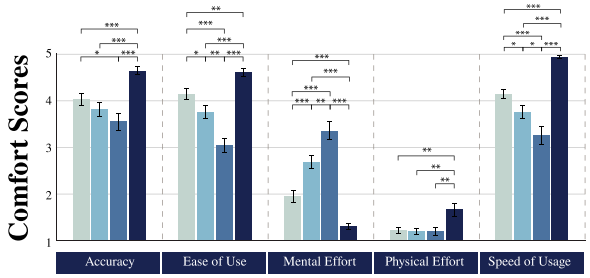
5.2.1. Sickness

The differences in terms of the sickness scores are negligible, as observed in Fig. 12. Across various categories, including *eye strain*, *headache*, *nausea*, and *overall sickness*, participants reported minimal



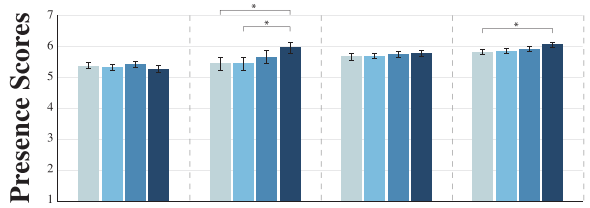
	Eye Strain	Headache	Nausea	Overall Sickness
Cartesian	1.17 ± 0.07	1.04 ± 0.03	1.29 ± 0.09	1.33 ± 0.08
Cylindrical	1.17 ± 0.07	1.04 ± 0.03	1.21 ± 0.07	1.33 ± 0.08
Spherical	1.19 ± 0.07	1.04 ± 0.03	1.25 ± 0.08	1.38 ± 0.08
Teleport	1.17 ± 0.07	1.04 ± 0.03	1.21 ± 0.07	1.19 ± 0.06

Fig. 12. Average sickness scores with ± SE. Sickness values are consistently low across all systems.



	Accuracy	Ease of Use	Mental Effort	Physical Effort	Speed of Usage
Cartesian	4.02 ± 0.13	4.15 ± 0.12	1.94 ± 0.13	1.21 ± 0.08	4.14 ± 0.10
Cylindrical	3.81 ± 0.15	3.75 ± 0.14	2.69 ± 0.14	1.19 ± 0.07	3.75 ± 0.13
Spherical	3.54 ± 0.17	3.04 ± 0.14	3.35 ± 0.18	1.19 ± 0.07	3.25 ± 0.19
Teleport	4.65 ± 0.08	4.60 ± 0.08	1.29 ± 0.07	1.65 ± 0.14	4.94 ± 0.04

Fig. 13. Average comfort scores with ± SE. The *Cartesian* system exhibited superior scores across multiple categories.



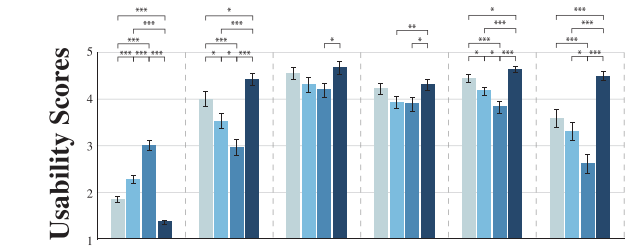
	Experienced Realism	General Presence	Involvement	Spatial Presence
Cartesian	5.36 ± 0.08	5.48 ± 0.17	5.68 ± 0.07	5.87 ± 0.06
Cylindrical	5.33 ± 0.08	5.48 ± 0.16	5.69 ± 0.07	5.89 ± 0.06
Spherical	5.40 ± 0.08	5.65 ± 0.16	5.72 ± 0.07	5.94 ± 0.06
Teleport	5.29 ± 0.08	5.98 ± 0.15	5.76 ± 0.07	6.06 ± 0.06

Fig. 14. Average presence scores with ± SE. All systems demonstrate a high level of presence.

differences and did not express significant complaints while using the system. The *overall sickness* score (1.30) indicates a low level of discomfort. Likewise, the scores for *headache* (1.04), *eye strain* (1.17), and *nausea* (1.24) indicate no substantial disparities between the systems.

5.2.2. Comfort

The comfort results, as shown in Fig. 13, reveal notable differences across the various systems. In terms of *ease of use*, the *Cartesian* system received relatively high ratings (4.15), whereas the *cylindrical* (3.75) and *spherical* (3.04) systems were perceived as less comfortable for voice-based navigation. The *teleportation* approach achieved the highest ease-of-use score at 4.60. These differences are mirrored in the *mental effort* ratings. The *spherical* system demanded the most cognitive load (3.35), followed by the *cylindrical* (2.69) and *Cartesian* (1.94) systems. The *teleportation* method required the least mental effort, scoring 1.29. Participants also reported varying perceptions of *accuracy* across the systems. The *Cartesian* system was rated highest among



	Complexity	Confidence	Consistency	Integration	Learnability	Reusability
Cartesian	1.83 ± 0.08	3.98 ± 0.13	4.54 ± 0.13	4.21 ± 0.13	4.43 ± 0.07	3.58 ± 0.18
Cylindrical	2.28 ± 0.10	3.52 ± 0.15	4.29 ± 0.15	3.92 ± 0.13	4.16 ± 0.09	3.29 ± 0.19
Spherical	3.00 ± 0.11	2.96 ± 0.17	4.19 ± 0.14	3.88 ± 0.15	3.82 ± 0.11	2.60 ± 0.20
Teleport	1.35 ± 0.05	4.42 ± 0.12	4.67 ± 0.12	4.29 ± 0.12	4.63 ± 0.06	4.48 ± 0.10

Fig. 15. Average usability scores with ± SE. The *Cartesian* system exhibited superior scores across multiple categories.

the voice-based techniques (4.02), followed by the *cylindrical* (3.81) and *spherical* (3.65) systems. Again, *teleportation* received the highest overall accuracy rating at 4.65. A similar pattern emerged for *speed of usage*. Participants rated the *Cartesian* system as the fastest among the voice-based techniques (4.14), followed by the *cylindrical* (3.75) and *spherical* (3.25) systems. *Teleportation* achieved the highest speed rating with 4.94. Regarding *physical effort*, differences between the systems were minimal, with an average score of 1.31 across the voice-based approaches. The only technique showing a slightly elevated physical demand was the *teleportation* method, which scored 1.65.

5.2.3. Presence

The results for presence, as illustrated in Fig. 14, reveal only minor differences between the systems across all measured categories. The average scores for *general presence* (5.64), *spatial presence* (5.94), *involvement* (5.71), and *experienced realism* (5.35) suggest a comparable sense of presence across all locomotion techniques. The *teleportation* approach showed slightly higher ratings in the categories of *general presence* (5.98) and *spatial presence* (6.06).

5.2.4. Usability

The usability scores, presented in Fig. 15, highlight both substantial and minor differences among the evaluated locomotion systems. The most prominent variations emerged in the categories of *reusability*, *confidence*, and *complexity*. Participants showed a stronger inclination to reuse the *Cartesian* system (3.58), whereas the *cylindrical* (3.29) and the *spherical* system (2.60) were less favored. The *teleportation* technique received the highest reusability score at 4.48. These preferences may be attributed, in part, to perceived differences in *complexity*. The *Cartesian* (1.83) and *cylindrical* (2.28) systems were rated as relatively simple to use, while the *spherical* system was perceived as more complex (3.00). *Teleportation* was considered the least complex, with a score of 1.35. Minor differences were observed in system *integration*, with *teleportation* (4.29) and the *Cartesian* system (4.21) slightly outperforming the *cylindrical* (3.92) and *spherical* (3.88) systems. A similar trend was found in *consistency*, where *teleportation* (4.67) and *Cartesian* navigation (4.54) received the highest ratings, followed by the *cylindrical* (4.29) and *spherical* (4.19) approaches. In terms of *learnability*, participants found the *teleportation* (4.64) and *Cartesian* (4.43) systems easier to learn compared to the *cylindrical* (4.16) and *spherical* (3.82) systems. Participants reported varying levels of *confidence* for each technique. The highest confidence was expressed for *teleportation* (4.67), followed by the *Cartesian* system (3.98), the *cylindrical* system (3.52), and finally the *spherical* system (2.96).

6. Discussion and future work

The following section addresses Research Questions 1–4 (see Section 4.1), discusses potential directions for future work, and outlines the inherent limitations of our system. Research Questions 1–2 focus exclusively on evaluating the performance of our voice-based approaches, while Research Question 3-5 compares their performance to the non-voice-based baseline of *teleportation*.

6.1. [Rq1] – Locomotion tasks

In our study, we investigated two different approaches to hands-free voice interaction within a digital environment: self-locomotion of the user and the locomotion or manipulation of external objects. This distinction was made to assess whether these systems have different application areas. It is important to treat these tasks separately and avoid comparing subjective measures between them.

However, our objective measures indicate that, within the same coordinate system, self-locomotion and object manipulation tasks produced very similar scores across most categories. Despite the similarity, self-locomotion consistently received slightly higher scores in every category. We also found that none of the coordinate systems or tasks induced any form of sickness during use. Furthermore, all systems received high presence scores, with no notable differences between them. Applying the framework of Melo et al. [86] to our presence results, all techniques lie within the acceptable range and achieved a grade of **B** for *experienced realism*, **B** for *spatial presence*, **A** for *involvement*, and **A** for *general presence*. Thus, both systems passed critical assessments and can be incorporated into any VR application without posing health risks or compromising immersion.

When comparing our results to similar voice-based work by Hombeck et al. [3,72], we found that most metrics, including sickness and presence scores, align closely with those observed in our system. However, slight differences emerged in terms of mental effort and complexity. These factors were slightly worse in our work compared to voice-based navigation systems that include landmarks. However, when compared to voice user interfaces for data interaction, our work showed better results. It is important to note that the work of Hombeck et al. [3] only included a 2D navigation task. The reduced mental effort observed in their results may stem from the simpler phrasing required to reach the destination, but it could also be influenced by the lower task complexity. In comparison to voice-based interaction with data, our system demonstrated higher user involvement scores, with participants reporting a stronger sense of immersion and realism. While the accuracy for other voice-based systems were similar, the speed of usage in our system showed an improvement, likely due to advancements in the voice system itself.

During the feedback sessions, we found that most participants perceived moving themselves as slightly more convenient than moving objects. However, they acknowledged the benefits of both systems, especially in scenarios where an alternative method of interaction is required. Thus, the *Cartesian*, *cylindrical*, and *spherical* locomotion techniques did not exhibit notable differences when applied to either self-locomotion or object manipulation, indicating their suitability for both tasks. However, significant differences were observed between the *Cartesian*, *cylindrical*, and *spherical* approaches themselves.

6.2. [Rq2] – Coordinate system

To evaluate the comparison between the *Cartesian*, *cylindrical*, and *spherical* coordinate systems, we employed multiple subjective and objective measures. Across self-locomotion as well as object manipulation approaches, the *Cartesian* system was consistently found to be more accurate, faster, and more user-friendly. While all systems performed similarly in object manipulation tasks, a clear preference emerged during self-locomotion tasks. Out of the 24 participants, 16 favored the *Cartesian* system, appreciating its simplicity, particularly the use of a single number for referencing positions instead of a combination of angles and distances. Eight participants preferred the *cylindrical* system, while none favored the *spherical* approach. This preference was further reflected in usability questionnaire responses, where the *Cartesian* system scored higher in categories such as *reusability*, *complexity*, *learnability*, and *confidence*. Similarly, it outperformed the *cylindrical* and *spherical* systems in terms of *ease of use*, *mental effort*, and *accuracy*. These results were largely influenced by the simpler positional encoding used in the *Cartesian* system, a point emphasized in feedback from nearly all participants. During the study, we also observed that some participants became confused about the distinction between longitude and latitude angles when using the spherical approach. This occasionally led to unintended, though correctly interpreted, locomotion commands. Although this issue was limited to a few participants, it may help explain the higher reported mental effort associated with the spherical coordinate system. Furthermore, in the Cartesian system, the encoding was continuously displayed on the ground, whereas in the cylindrical and spherical systems, only guiding lines were visible and the exact position had to be estimated. This difference may have influenced the perceived mental effort. However, the error in the object manipulation task was similar across all voice-based conditions, suggesting that even when participants had to estimate the target position, the resulting accuracy was comparable to conditions in which numerical values were continuously displayed.

While we observe differences in mental effort between the techniques, it is important to note that, in the current setup, participants had their hands free and were not engaged in additional tasks, such as moving objects or performing intricate fine motor activities. Performing secondary tasks would likely further increase the mental complexity, making it even more critical to minimize the cognitive load associated with locomotion. Although the precise impact of secondary tasks on mental effort requires further investigation, it should be carefully considered when selecting an appropriate locomotion technique, further highlighting the advantages of the *Cartesian* encoding.

While the number of commands used to complete a task was quite similar, notable differences in task completion time emerged. Although the *spherical* system was designed for greater precision, its increased complexity ultimately resulted in similar or even slightly worse accuracy, along with longer completion times in practice, compared to the more intuitive *Cartesian* approach. This can again be attributed to the more complex position encoding required by the *spherical* system. The goal of creating a system that combined elements of both the *Cartesian* and *spherical* approaches resulted in a system whose performance consistently fell between the two. The *cylindrical* system generally performed better than the *spherical* approach across most tasks but remained less effective than the *Cartesian* system. Although it was favored by some participants and consistently outperformed the *spherical* approach, it was never able to surpass the *Cartesian* system in overall performance.

Therefore, we would strongly advocate for the *Cartesian* encoding method. This approach offers faster completion times while maintaining the same level of accuracy and the same number of required voice commands.

6.3. [Rq3] – Interaction modality

To compare the concept of voice-based locomotion against a baseline VR locomotion technique, we integrated *teleportation* into our study. While the *teleportation* approach demonstrated higher performance across most categories, voice-based systems proved capable of approximating the performance of non-voice-based systems, with a few areas still needing improvement. Particularly, notable differences emerged in areas such as mental effort, complexity, accuracy, and reusability. These differences were especially pronounced when comparing teleportation to the *spherical* system, while the gap between

teleportation and the *Cartesian* system was significantly smaller. However, in key aspects critical to VR locomotion, such as motion sickness and presence scores, there were only minor to negligible differences between the interaction modalities. One major difference observed during our study was that the teleportation technique struggled to overcome the lower and upper barricades in our maze. Since teleportation requires a clear line of sight to the destination, blocked vision made it considerably more cumbersome to reach the target location. For the lower barricade, users typically teleported close to the wall and stretched their arms upward to point to the other side. In contrast, for the upper barricade, users often crouched down in an attempt to aim beyond the obstacle. Voice-based systems, on the other hand, allowed participants to simply articulate the number of the target location on the other side of the wall, enabling them to overcome such obstacles effortlessly and without the need for additional physical adjustments. This difference likely explains why the teleportation technique required more commands to complete the maze task compared to the voice-based systems.

With this in mind, we can generally conclude that while voice-based systems can still be further refined to more closely match approaches like *teleportation*, they already offer unique opportunities. Especially in scenarios where controllers are infeasible and techniques like teleportation are not available, voice-based locomotion, particularly using the *Cartesian* system, emerges as a viable alternative. Although it does not yet perfectly match the performance of the *teleportation* approach, it demonstrates strong potential to closely approximate the baseline.

6.4. [Rq4] – Interaction patterns

To explore focus areas during hands-free locomotion, we closely examined the interaction patterns associated with each task and system. The maze setup was intentionally designed to limit participants' field of view, featuring narrow corridors that restricted visibility. In contrast, the platform environment offered wide areas with free movement. Despite these differences, both the restricted and unrestricted configuration showed a clear pattern: participants primarily interacted with areas near their central view direction, as illustrated in Figs. 8, 10, 9 and 11. While keeping the navigation within the field of view was expected to some degree, we observed a consistent alignment of the navigation direction with the user's central view across all techniques. This pattern was not limited to the voice-based approaches. It was also evident in the baseline teleportation technique, where users rarely deviated from their forward gaze, despite having the flexibility to do so easily. Therefore, we believe this behavior across all techniques is driven more by user preference rather than by constraints imposed by the locomotion design. A slight bias towards the center may be attributed to the design of the *cylindrical* and *spherical* approaches, both of which define 0 degrees directly at the center of the user's field of view. In contrast, the *Cartesian* approach displays all values simultaneously, enabling users to more easily deviate from the central rows. In further contrast, the *teleportation* system lacks a defined central reference entirely, providing users with complete freedom to select any destination within the environment.

These findings suggest a strong alignment between visual attention and interaction zones, even when participants were given greater freedom of movement. We therefore recommend focusing primarily on the central field of view when developing locomotion techniques to avoid cluttering the virtual space with unnecessary information. The specific design and implementation of these techniques remain an area for future research.

6.5. [Rq5] – Voice locomotion

To address the overarching question of whether voice commands can support locomotion in areas where landmarks are absent, we have considered the discussion of Research Questions [Rq1] - [Rq4], as well as objective and subjective measures and user feedback. Based on this analysis, we can confidently assert that voice commands can serve as a viable alternative for navigation in scenarios where the use of physical controllers is impractical or infeasible. While non-voice-based approaches demonstrate higher accuracy, faster time-to-target, and better overall usability, voice-based systems still receive positive evaluations across all questionnaire metrics and can compete with the performance of non-voice-based systems. The most significant difference lies in the time-to-target, where the *teleportation* approach remains substantially faster. However, the voice-based approaches enhance system accessibility, allowing individuals who might otherwise struggle to interact with the environment to do so more effectively. It also supports both self-locomotion and object manipulation in environments where physical controllers are infeasible or impractical. An interesting direction for future work would be to investigate whether the post-study ratings correspond with physiological measurements such as eye tracking, heart rate, or skin conductance. Such data could further provide objective insights into user engagement, cognitive load, and emotional responses during interaction. While the *Cartesian*, *cylindrical*, and *spherical* systems performed similarly in terms of overall performance and accuracy, we observed a clear advantage in time to target and user satisfaction with the *Cartesian* system. We attribute this improvement to its simpler position encoding. Consequently, we recommend adopting a more straightforward encoding method, as it significantly enhances usability. The insights into interaction patterns can further guide the development of specialized locomotion systems, enabling more intuitive and efficient navigation in virtual environments. By aligning locomotion systems with user behavior and preferences, we can enhance the overall effectiveness and user experience in hands-free interaction.

6.6. Limitations

While the presented coordinate systems were perceived positively, our approach does pose some inherent limitations. For example, the coordinate systems impose different cognitive loads. The *Cartesian* system enables users to specify a desired location with a single number, the *cylindrical* system requires two numbers, whereas the *spherical* approach often necessitates specifying two angles and a distance, resulting in a higher cognitive demand. Although our intent was to investigate these cognitive differences, we acknowledge the inherent limitations imposed by cognitive disparities. Furthermore, our approach focused exclusively on user-centered coordinate systems. While the presented systems could also be applied as world-space coordinate systems, their performance in this context will need to be evaluated in future work. An additional limitation of our approach may be the range of visible numbers. While in our study even the most distant numbers remained clearly visible to participants, scaling this approach to larger environments or greater distances may present challenges. In such cases, the visibility of the numbers displayed on the ground would need to be carefully considered and potentially adapted to ensure continued usability. Furthermore, each participant spent approximately 15 min using each technique, which is relatively brief for evaluating long-term usability. Once a suitable technique is identified, further investigation should be conducted to assess how users perform with these techniques during more extended usage sessions. Additionally, the study was conducted under controlled laboratory conditions. The integration of voice-based systems into everyday workflows is beyond the scope of this work and should be explored in future research. This also applies to the intended target group. In our study, most participants came from fields such as information technology or mathematics. As a result, working with coordinate systems may have been easier for them compared to the general population. An in-depth analysis involving diverse target groups is planned for future research.

7. Conclusion

Traditional input methods, such as physical controllers, may not always be optimal and can, in some cases, even be impractical [3]. In such cases, voice-controlled approaches can offer an effective alternative for interacting with the environment. However, articulating a desired locomotion destination can be challenging in spaces without discernible landmarks, such as empty rooms, or when multiple instances of the same object are present, as in a forest filled with numerous trees.

We address this challenge by facilitating hands-free locomotion and interaction in landmark-free environments through the development of a system that visually aids in selecting a location. We therefore compare *Cartesian*, *cylindrical*, and *spherical* coordinate systems approaches to the non-voice-based approach of teleportation. We evaluate all systems using both objective and subjective measures to assess performance, accuracy, and user satisfaction. We found that while all voice-based systems performed similarly in accuracy, the *Cartesian* and *cylindrical* systems resulted in faster task completion and higher user satisfaction due to their simpler position encoding. While our non-voice-based baseline demonstrated slightly higher performance in most categories, the voice-based systems, particularly the *Cartesian* approach, came very close in terms of both performance and user satisfaction. Consequently, in environments where controllers are unavailable or impractical, voice-based systems offer a viable and effective alternative for interacting with digital environments, even in the absence of landmarks.

Furthermore, we observed that participants consistently interacted with areas close to their central view direction, regardless of the system used or the degree of freedom provided. This interaction pattern highlights the importance of designing systems that align with natural user behavior, which is crucial for developing more intuitive and accessible interactive systems in virtual reality and other digital environments. While the performance of the different systems is relatively similar, personal preferences and subjective evaluations can vary across techniques. This underscores the importance of usability and intuitiveness, which are critical factors for the successful adoption of new interaction methods.

Our work advances the broader field of virtual reality by enhancing accessibility and user experience, moving us closer to more intuitive and inclusive interactive systems. It specifically advances the field of voice-controlled interaction by offering a solution to the challenge of articulating position commands in environments lacking discernible landmarks.

CRediT authorship contribution statement

Jan Hombeck: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Henrik Voigt:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Conceptualization. **Kai Lawonn:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Acknowledgments

This work was supported by the Carl Zeiss Foundation within the project Sensorized Surgery, Germany (P2022-06-004).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cag.2025.104318.

Data availability

Data will be made available on request.

References

[1] Lange M, Hjalmarsson J, Cooper M, Ynnerman A, Duong V. 3D visualization and 3d and voice interaction in air traffic management. In: The annual SIGRAD conference. special theme-real-time simulations. Citeseer; 2003, p. 17–22.
[2] Sin J, Munteanu C. Let's go there: Voice and pointing together in VR. In: 22nd International Conference on Human-Computer Interaction with Mobile Devices and Services. 2020.
[3] Hombeck J, Voigt H, Heggemann T, Datta RR, Lawonn K. Tell me where to go: Voice-controlled hands-free locomotion for virtual reality systems. In: IEEE VR. 2023, p. 123–34.
[4] Brown T, Mann B, Ryder N, Subbiah M, Kaplan JD, Dhariwal P, et al. Language models are few-shot learners. Adv Neural Inf Process Syst 2020;33:1877–901.
[5] Radford A, Kim JW, Xu T, Brockman G, McLeavey C, Sutskever I. Robust speech recognition via large-scale weak supervision. 2022, arXiv preprint arXiv:2212.04356.
[6] Calandra D, Pratticò FG, Lamberti F. Comparison of hands-free speech-based navigation techniques for virtual reality training. In: MELECON. IEEE; 2022, p. 85–90.
[7] Monteiro P, Gonçalves G, Coelho H, Melo M, Bessa M. Hands-free interaction in immersive virtual reality: A systematic review. IEEE TVCG 2021;27:2702–13.
[8] Mewes A, Hensen B, Wacker F, Hansen C. Touchless interaction with software in interventional radiology and surgery: a systematic literature review. Int J Comput Assist Radiol Surg 2017;12(2):291–305.
[9] Cronin S, Doherty G. Touchless computer interfaces in hospitals: A review. Heal Inform. J 2019;25(4):1325–42.
[10] Wang P, Wu P, Wang J, Chi H-L, Wang X. A critical review of the use of virtual reality in construction engineering education and training. Environ Res Public Heal 2018;15.
[11] Ferracani A, Faustino M, Giannini GX, Landucci L, Bimbo A. Natural experiences in museums through virtual reality and voice commands. ACM; 2017.
[12] Lugrin J-L, Kern F, Schmidt R, Kleinbeck C, Roth D, Daxer C, et al. A location-based VR museum. In: 2018 10th International Conference on Vir- tual Worlds and Games for Serious Applications (VS-Games). 2018, p. 1–2.
[13] Yoshinaga T, Arita D, Miyazaki W, Masuda K. Development of tele-echography interface with AR/VR visualization system of internal organs. IEEE BioCAS 2011;90–3.
[14] Pajorová E, Hluchý L, Kostic I, Pajorova J, Bacáková M, Zatloukal M. A virtual reality visualization tool for three-dimensional biomedical nanostructures. J Phys 2018;1098.
[15] Nebylov AV, Watson J. Aerospace navigation systems. John Wiley & Sons; 2016.
[16] Martinez ES, Wu AS, McMahan RP. Research trends in virtual reality locomotion techniques. In: 2022 IEEE VR. IEEE; 2022, p. 270–80.
[17] Bowman DA, Kruijff E, LaViola Jr JJ, Poupyrev I. 3D user interfaces: theory and practice. Addison-Wesley; 2005.
[18] Buttussi F, Chittaro L. Locomotion in place in virtual reality: A comparative evaluation of joystick, teleport, and leaning. IEEE TVCG 2019;27(1):125–36.
[19] Chen W, Plancoulaine A, Férey N, Touraine D, Nelson J, Bourdot P. 6Dof navigation in virtual worlds: comparison of joystick-based and head-controlled paradigms. In: VRST. 2013.
[20] Lugrin J-L, Juchno A, Schaper P, Landeck M, Latoschik ME. Drone-steering: A novel VR traveling technique. In: Proceedings of the 25th ACM Symposium on VR Software and Technology. 2019.
[21] Cho I, Li J, Wartell Z. Multi-scale 7DOF view adjustment. IEEE Vis Comput Graph 2017;24(3):1331–44.
[22] Raees M, Ullah S, Rahman SU. VEN-3DVE: vision based egocentric navigation for 3D virtual environments. Int J Interact Des Manuf 2019;13(1):35–45.
[23] Hassouna MS, Farag AA, Falk R. Virtual fly-over: A new visualization technique for virtual colonoscopy. MICCAI 2006;9:381.
[24] Paulo SF, Medeiros D, Borges PB, Jorge J, Lopes DS. Improving camera travel for immersive colonography. In: IEEE VRW. IEEE; 2020, p. 748–9.
[25] Langbehn E, Eichler T, Ghose S, von Luck K, Bruder G, Steinicke F. Evaluation of an omnidirectional walking-in-place user interface with virtual locomotion speed scaled by forward leaning angle. In: GI vR/AR. 2015.
[26] Bozgeyikli E, Raij A, Katkoori S, Dubey R. Locomotion in virtual reality for room scale tracked areas. Int J Hum-Comput Stud 2019;122:38–49.
[27] Habgood MJ, Moore D, Wilson D, Alapont S. Rapid, continuous movement between nodes as an accessible virtual reality locomotion technique. In: IEEE VR. IEEE; 2018, p. 371–8.

[28] Hombeck JN, Lichtenberg N, Lawonn K. Evaluation of spatial perception in virtual reality within a medical context. In: Bildverarbeitung für die medizin. Springer; 2019, p. 283–8.

[29] Hombeck J, Lichtenberg N, Lawonn K. Heads up a study of assistive visualizations for localisation guidance in virtual reality. In: BVM. Springer; 2022, p. 83–8.

[30] Hombeck J, Meuschke M, Zyla L, Heuser AJ, Toader J, Popp F, et al. Evaluating perceptional tasks for medicine: A comparative user study between a virtual reality and a desktop application. In: 2022 IEEE VR. 2022, p. 514–23.

[31] Hombeck J, Meuschke M, Lieb S, Lichtenberg N, Datta R, Krone M, et al. Distance visualizations for vascular structures in desktop and vr: Overview and implementation. In: VCBM. Eurographics; 2022, p. 129–335.

[32] Prithul A, Bhandari J, Spurgeon W, Folmer EO. Evaluation of hands-free teleportation in VR. ACM 2022.

[33] Chen Y, Granados DFP, Kadone H, Suzuki K. Control interface for hands-free navigation of standing mobility vehicles based on upper-body natural movements. IROS 2020;11322–9.

[34] Tregillus S, Folmer EO. VR-STEP: Walking-in-place using inertial sensing for hands free navigation in mobile VR environments. CHI Conf Hum Factors Comput Syst 2016.

[35] Klamka K, Siegel A, Vogt S, Göbel F, Stellmach S, Dachselt R. Look & Pedal: Hands-free navigation in zoomable information spaces through gaze-supported foot input. ACM; 2015.

[36] Tregillus S, Zayer MA, Folmer EO. Handsfree omnidirectional VR navigation using head tilt. In: CHI Conference on Human Factors in Computing Systems. 2017.

[37] Zielasko D, Horn S, Freitag S, Weyers B, Kuhlen TW. Evaluation of hands-free HMD-based navigation techniques for immersive data analysis. In: 2016 IEEE 3DUI. IEEE; 2016, p. 113–9.

[38] Flemming C, Weyers B, Zielasko D. How to take a brake from embodied locomotion–seamless status control methods for seated leaning interfaces. In: 2022 IEEE conference on virtual reality and 3D user interfaces. IEEE; 2022, p. 728–36.

[39] Cannavò A, Calandra D, Pratticò FG, Gatteschi V, Lamberti F. An evaluation testbed for locomotion in virtual reality. IEEE TVCG 2020;27(3):1871–89.

[40] Bowman DA, Koller D, Hodges LF. Travel in immersive virtual environments: An evaluation of viewpoint motion control techniques. In: International symposium on virtual reality. IEEE; 1997, p. 45–52.

[41] Loup G, Loup-Escande E. Effects of travel modes on performances and user comfort: a comparison between ArmSwinger and teleporting. Int J HCI 2019;35(14):1270–8.

[42] Calandra DM, Billi M, Lamberti F, Sanna A, Borchiellini R. Arm swinging vs treadmill: A comparison between two techniques for locomotion in virtual reality. In: Eurographics. 2018.

[43] Ferracani A, Pezzatini D, Bianchini J, Biscini G, Del Bimbo A. Locomotion by natural gestures for immersive virtual environments. In: AltMM. 2016, p. 21.

[44] Pai YS, Kunze K. Armswing: Using arm swings for accessible and immersive navigation in ar/vr spaces. In: Conference on mobile and ubiquitous multimedia. 2017, p. 189–98.

[45] Wilson G, McGill M, Jamieson M, Williamson JR, Brewster SA. Object manipulation in virtual reality under increasing levels of translational gain. In: Proceedings of the 2018 CHI conference on human factors in computing systems. 2018, p. 1–13.

[46] Calandra D, Lamberti F, Migliorini M. On the usability of consumer locomotion techniques in serious games: Comparing arm swinging, treadmills and walk-in-place. In: ICCE. IEEE; 2019, p. 348–52.

[47] Sarupuri B, Hoermann S, Whitton MC, Lindeman RW. LUTE: A locomotion usability test environmentfor virtual reality. In: 2018 10th international conference on virtual worlds and games for serious applications (VS-games). 2018, p. 1–4.

[48] Chebotarev V, Brezitskaya V, Kovalev I, Kartsan I, Malanina YN, Shemyakov A. Solving navigation-temporal tasks in different coordinate systems. In: IOP conference series: materials science and engineering. IOP Publishing; 2018, 022029.

[49] Biswas P, Robinson P. A new screen scanning system based on clustering screen objects. Assist Technol 2008;2(3):24–31.

[50] Blackstien-Adler S, Shein F, Quintal J, Birch S, Weiss PLT. Mouse manipulation through single-switch scanning. Assist Technol 2004;16(1):28–42.

[51] Pinheiro CG, Naves EL, Pino P, Losson E, Andrade AO, Bourhis G. Alternative communication systems for people with severe motor disabilities: a survey. BioMed Online 2011;10:1–28.

[52] Pasqualotto E, Matuz T, Federici S, Ruf CA, Bartl M, Olivetti Belardinelli M, et al. Usability and workload of access technology for people with severe motor impairment: a comparison of brain-computer interfacing and eye tracking. Neurorehabil Neural Repair 2015;29(10):950–7.

[53] Folmer E, Liu F, Ellis B. Navigating a 3D avatar using a single switch. In: Proceedings of the 6th international conference on foundations of digital games. 2011, p. 154–60.

[54] Surdilovic T, Zhang YQ. Convenient intelligent cursor control web systems for internet users with severe motor-impairments. Int J Med Inform 2006;75(1):86–100.

[55] Wang S, Liu B, Feng Y, Meng L. Mixed reality-based coordinate system for enhancing survey knowledge acquisition: a study on allocentric and egocentric perspectives. Cartogr Geogr Inf Sci 2025;52(2):160–80.

[56] Panayotov V, Chen G, Povey D, Khudanpur S. Librispeech: An ASR corpus based on public domain audio books. IEEE ICASSP 2015;5206–10.

[57] Baevski A, Zhou H, rahman Mohamed A, Auli M. Wav2vec 2.0: A framework for self-supervised learning of speech representations. 2020, arXiv. arXiv:2006.11477.

[58] Chaturvedi SS, Sailor HB, Patil HA. Noisy student teacher training with self supervised learning for children ASR. SPCOM 2022;1–5.

[59] Baevski A, Schneider S, Auli M. Vq-wav2vec: Self-supervised learning of discrete speech representations. 2019, arXiv.

[60] Wei J, Bosma M, Zhao VY, Guu K, Yu AW, Lester B, et al. Finetuned language models are zero-shot learners. 2021, arXiv preprint arXiv:2109.01652.

[61] Dey N, Gosal G, Khachane H, Marshall W, Pathria R, Tom M, et al. Cerebras-GPT: Open compute-optimal language models trained on the cerebras wafer-scale cluster. 2023, arXiv.

[62] Touvron H, Lavril T, Izacard G, Martinet X, Lachaux M-A, Lacroix T, et al. Llama: Open and efficient foundation language models. 2023, arXiv.

[63] Wang Y, Mishra S, Alipoormolabashi P, Kordi Y, Mirzaei A, Arunkumar A, et al. Super-NaturalInstructions:Generalization via declarative instructions on 1600+ tasks. In: EMNLP. 2022.

[64] Hinton G, Vinyals O, Dean J. Distilling the knowledge in a neural network. 2015, arXiv preprint arXiv:1503.02531.

[65] Frantar E, Ashkboos S, Hoefler T, Alistarh D. GPTQ: Accurate post-training quantization for generative pre-trained transformers. 2022, arXiv preprint arXiv:2210.17323.

[66] Onnx Runtime developers. ONNX runtime. 2023, Version: 1.14.0. https://www.onnxruntime.ai.

[67] Rantamaa HR, Kangas J, Jordan M, Mehtonen H, Mäkelä J, Ronkainen K, et al. Evaluation of voice commands for mode change in virtual reality implant planning procedure. Int J Comput Assist Radiol Surg 2022;17:1981–9.

[68] Kern F, Keser T, Niebling F, Latoschik ME. Using hand tracking and voice commands to physically align virtual surfaces in AR for handwriting and sketching with HoloLens 2. In: Proceedings of the 27th ACM Symposium on VR Software and Technology. 2021.

[69] Ferreira J, Mendes D, Nóbrega R, Rodrigues RA. Immersive multimodal and procedurally-assisted creation of vr environments. 2021 IEEE VRW 2021;30–7.

[70] Chilufya EM, Arvola M. Conceptual designing of a virtual receptionist: Remote desktop walkthrough and bodystorming in VR. In: 9th International Conference on Human-Agent Interaction. 2021.

[71] Johnsen K, Dickerson RF, Raij A, Lok BC, Jackson J, Shin M, et al. Experiences in using immersive virtual characters to educate medical communication skills. IEEE VR 2005. 2005;179–86.

[72] Hombeck J, Voigt H, Lawonn K. Voice user interfaces for effortless navigation in medical virtual reality environments. Comput Graph 2024.

[73] Sweller J. Cognitive load during problem solving: Effects on learning. Cogn Sci 1988;12(2):257–85.

[74] Mayer RE, Moreno R. Nine ways to reduce cognitive load in multimedia learning. Educ Psychol 2003;38(1):43–52.

[75] Mine MR, Brooks Jr FP, Sequin CH. Moving objects in space: exploiting proprioception in virtual-environment interaction. In: Proceedings of the 24th annual conference on computer graphics and interactive techniques. 1997, p. 19–26.

[76] LaViola Jr. JJ, Kruijff E, McMahan RP, Bowman D, Poupyrev IP. 3D user interfaces: theory and practice. Addison-Wesley Professional; 2017.

[77] Haas JK. A history of the unity game engine. Diss Worcester Polytechnic Institute, Worcester Polytechnic Institute; 2014.

[78] Corradini A, Cohen PR. On the relationships among speech, gestures, and object manipulation in virtual environments: Initial evidence. Adv Nat Multimodal Dialogue Syst 2005;97–112.

[79] Williams AS, Garcia J, Ortega F. Understanding multimodal user gesture and speech behavior for object manipulation in augmented reality using elicitation. IEEE Trans Vis Comput Graphics 2020;26(12):3479–89.

[80] Sarupuri B, Jung S, Hoermann S, Whitton MC, Lindeman RW. Testbed evaluation of multi-travel mode in virtual reality. In: VRST. 2020.

[81] Schuemie M, Abel B, Van Der Mast C, Krijn M, Emmelkamp P. The effect of locomotion technique on presence, fear and usability in a virtual environment. In: Euromedia, vol. 2005, Citeseer; 2005, p. 11th.

[82] Bimberg P, Weissker T, Kulik A. On the usage of the simulator sickness questionnaire for virtual reality research. In: VRW 2020. IEEE; 2020, p. 464–7.
[83] Kennedy RS, Lane NE, Berbaum KS, Lilienthal MG. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. Aviat Psychol 1993;3(3):203–20.
[84] Douglas SA, Kirkpatrick AE, MacKenzie IS. Testing pointing device performance and user assessment with the iso 9241, part 9 standard. In: SIGCHI. 1999, p. 215–22.
[85] Schubert T, Friedmann F, Regenbrecht H. The experience of presence: Factor analytic insights. Presence: Teleoperators & Virtual Environ 2001;10(3):266–81.
[86] Melo M, Gonçalves G, Bessa M, et al. How much presence is enough? qualitative scales for interpreting the igroup presence questionnaire score. IEEE Access 2023;11:24675–85.
[87] Brooke J, et al. SUS-a quick and dirty usability scale. Usability Eval Ind 1996;189(194):4–7.